

Fish Scale Derived Biochar: a Sustainable Adsorbent for Simultaneous Heavy Metal Removal from Contaminated Water

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Background

- Heavy metals (Pb, Cr, Cd, Ni) are toxic, non-biodegradable pollutants commonly found in industrial wastewater.
- Conventional treatment methods are often expensive, energy-intensive, and generate secondary pollution.
- Biochar is a low-cost, porous, carbon-rich material with high adsorption capacity for heavy metals.
- Fish scales, a major waste product from the seafood industry, are rich in calcium phosphate and collagen, ideal precursors for biochar.

Motivation

- Repurposing fish scale waste reduces environmental burden and supports circular economy practices.
- Fish scale-derived biochar may offer a sustainable, low-cost alternative to synthetic adsorbents.
- Developing efficient treatment methods is critical for clean water access, especially in low-resource areas.



Objectives

- To synthesize and characterize biochar derived from fish scales.
- To evaluate its adsorption efficiency for multiple heavy metals (Pb, Cd, Cr, Ni) in contaminated water.
- To optimize adsorption conditions (pH, contact time, temperature, biochar dosage).
- To assess reusability and compare performance with commercial adsorbents.

Table 1. ICP-OES analytic conditions.

Parameter	Value
Selected Wavelengths	Cd 228.502; Cr 267.716; Ni 231.604; and Pb 220.353
Plasma view	Axial
Gas flows	Plasma (14 L/min), auxiliary (0.2 L/min), and nebulizer (0.7 L/min)
RF power	1350 W
Read delay	60 s
Replicates	3
Peristaltic pump flow rate	1.5 mL/min

Methodology

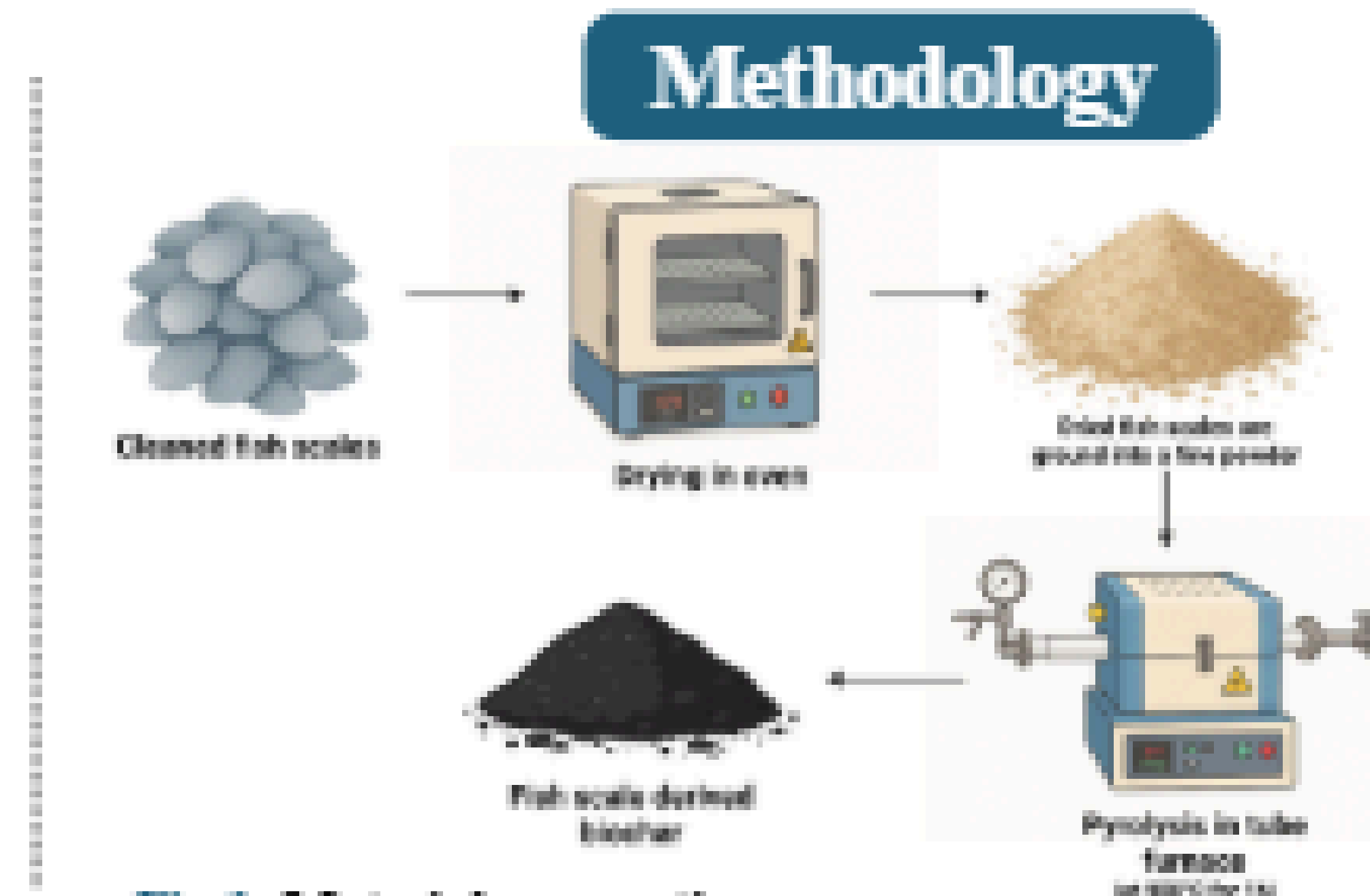


Fig 1. Material preparation.



Fig 2. Method optimization.

Material Characterization

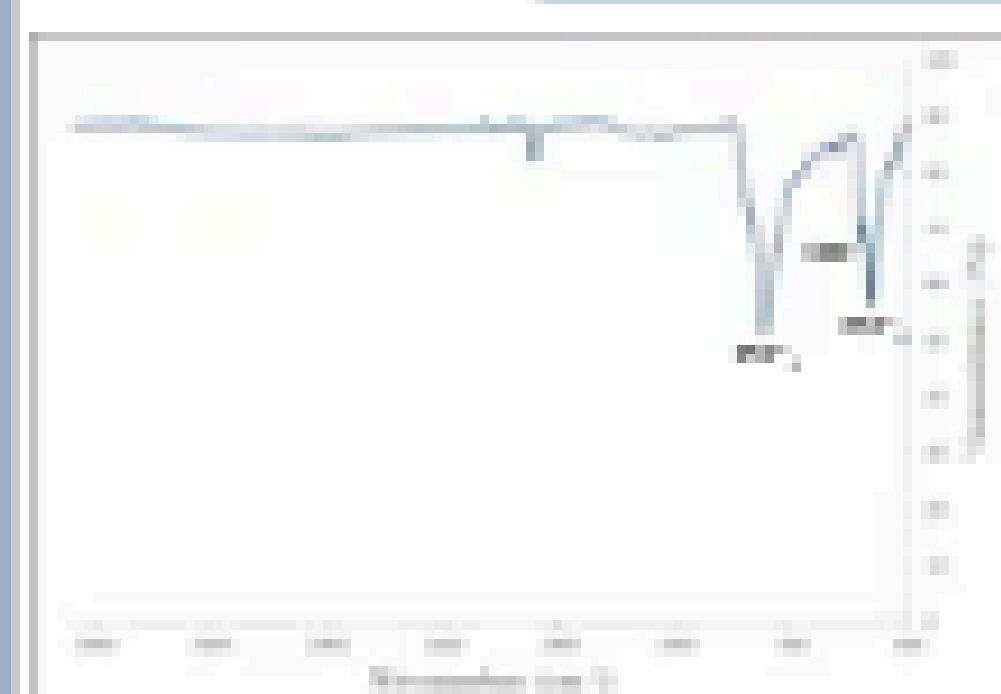


Fig 1. FTIR of FSBC.

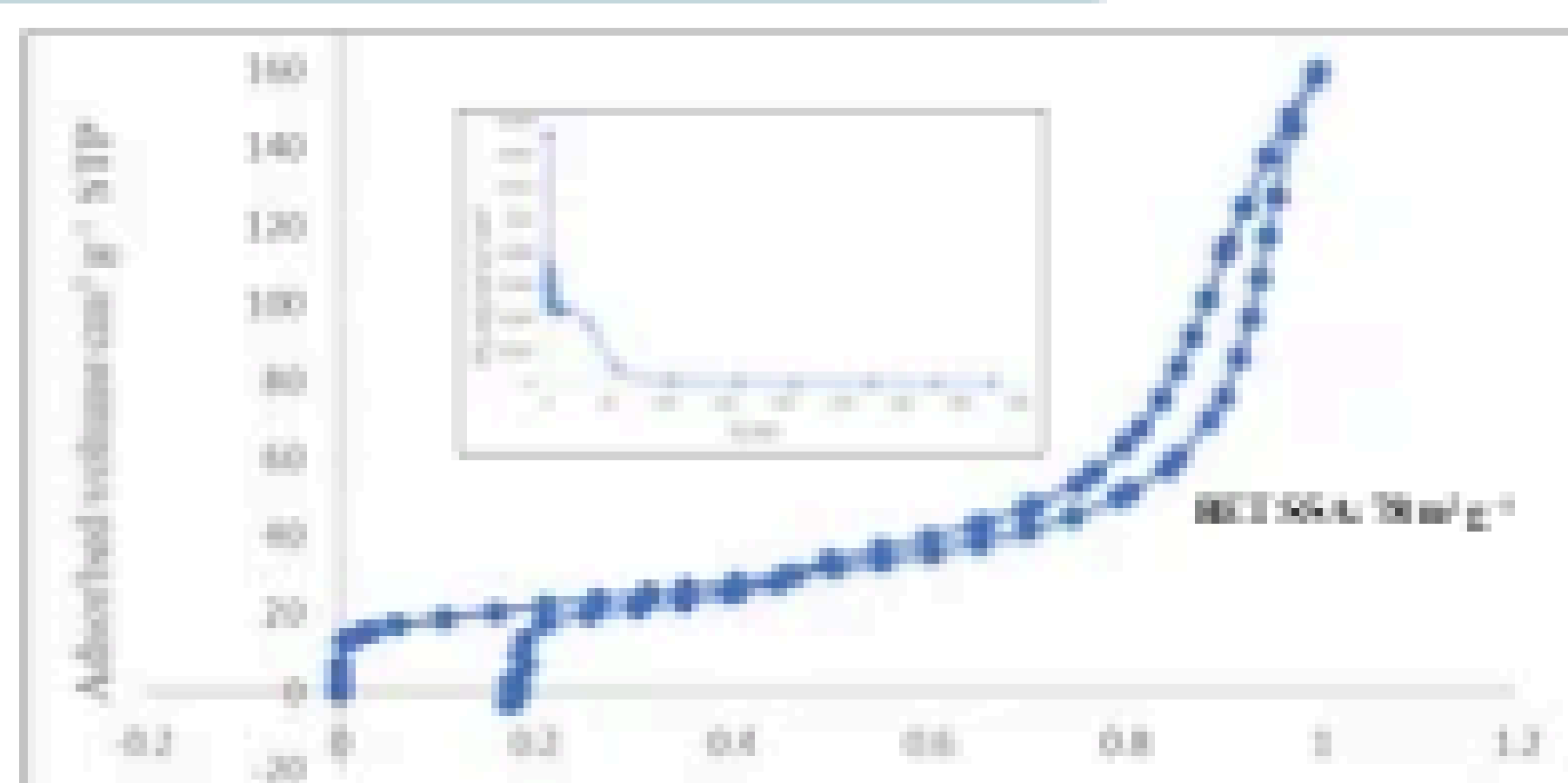


Fig 4. Pure size distribution of FSBC, shown to be monodisperse, and No adsorption-desorption analysis isotherm.



Fig 3. SEM image of FSBC.

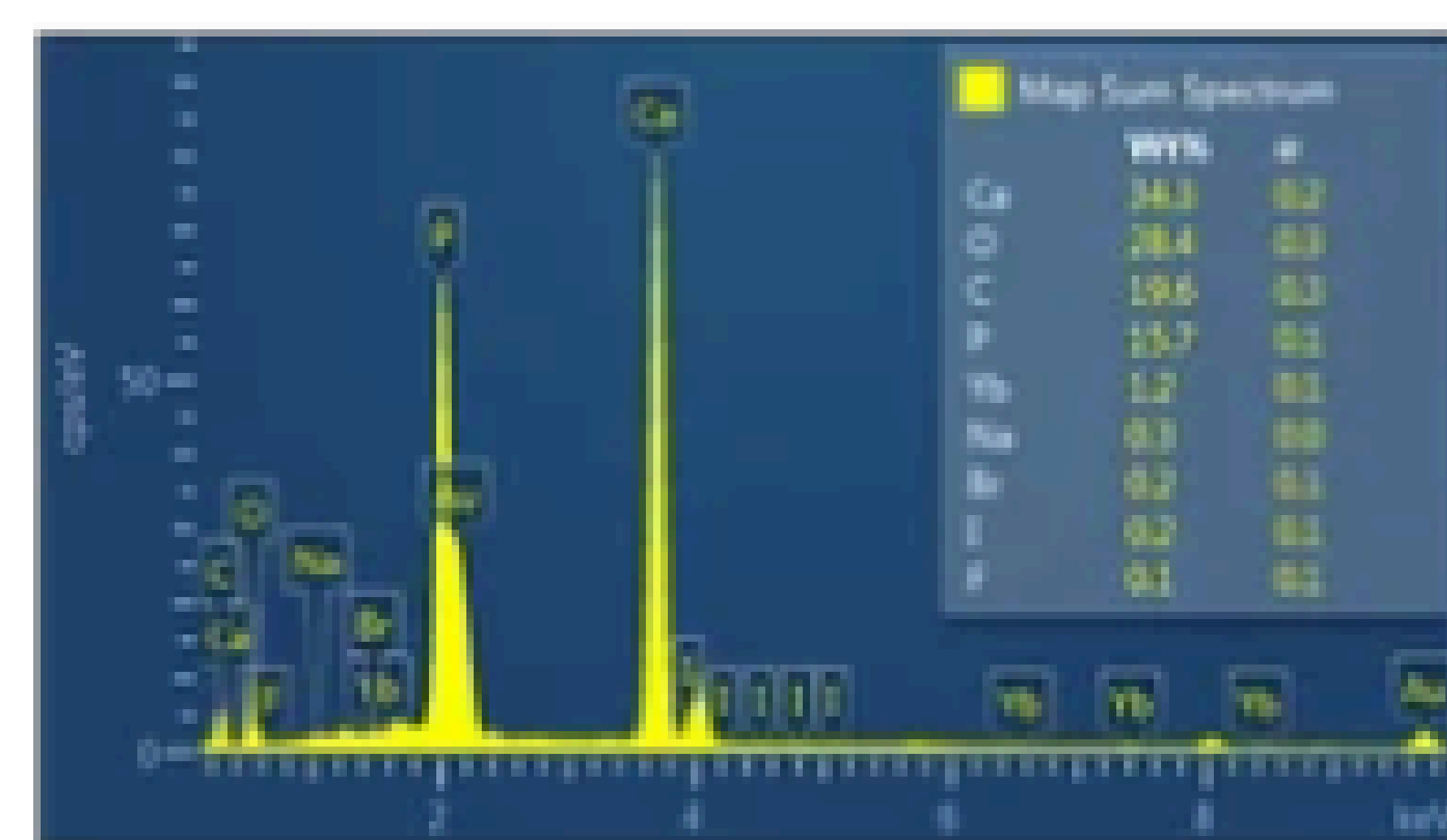


Fig 5. EDX spectra of FSBC

Optimization

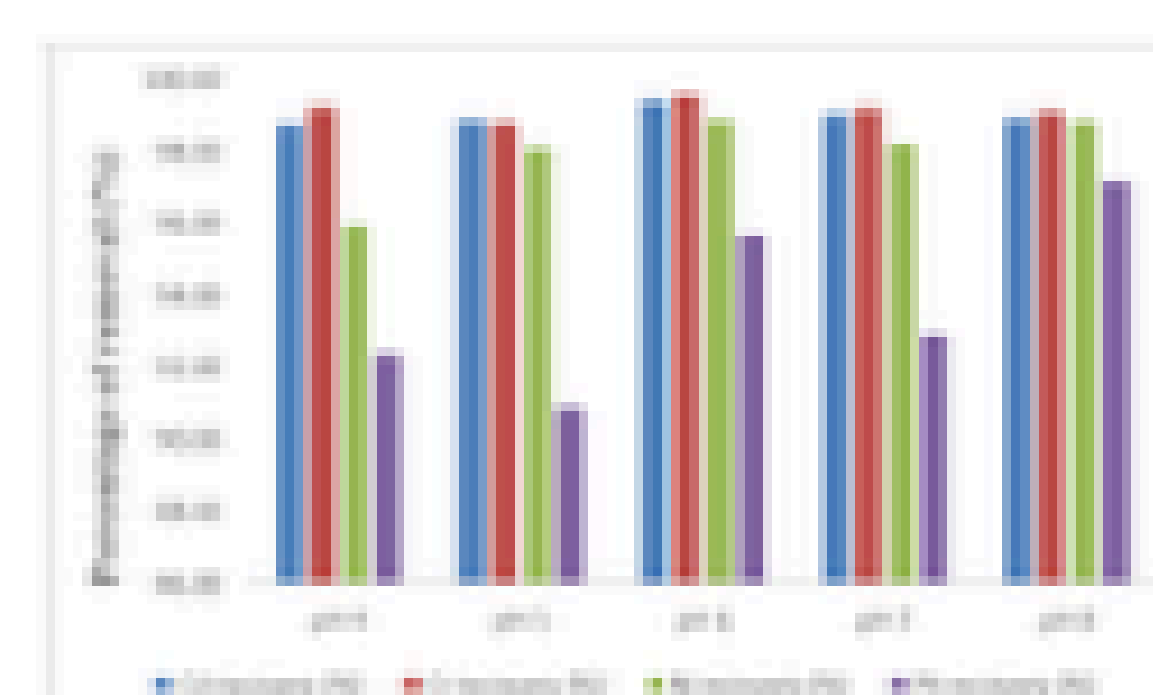


Fig 8. Effect of pH on Cd, Cr, Ni, and Pb (200ppb) removal from 25 ml aqueous solution using FSBC (25 mg) over 1h.

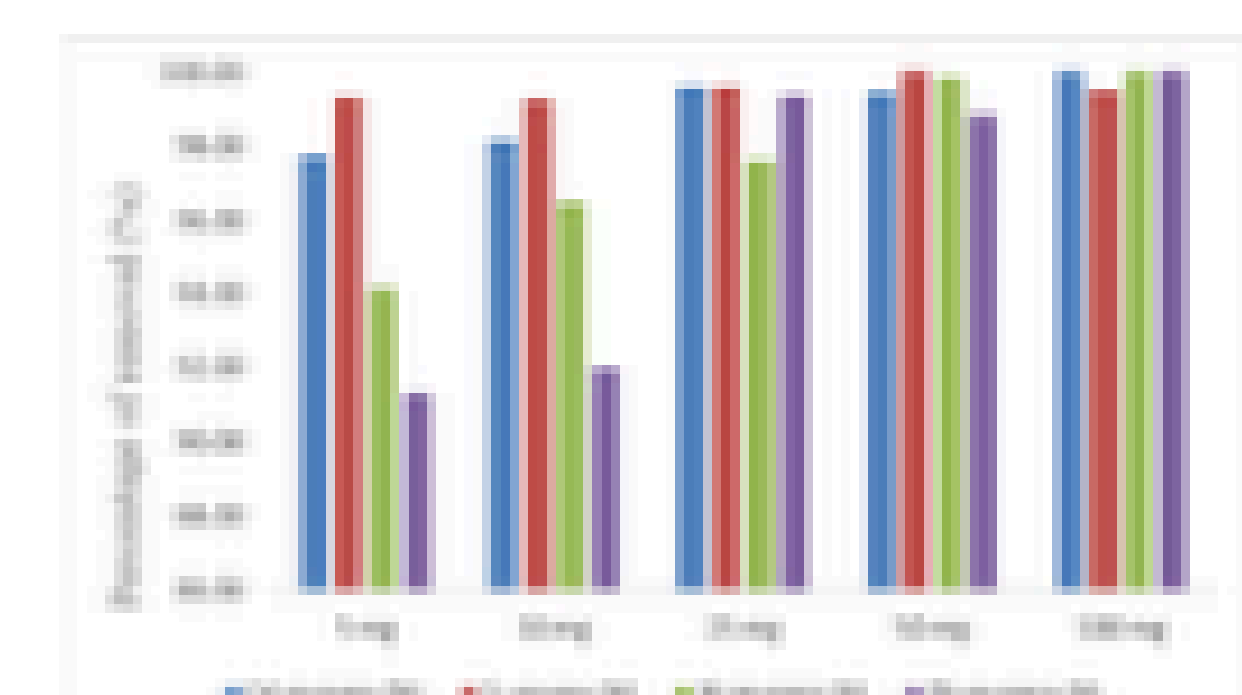


Fig 9. Effect of dosage on Cd, Cr, Ni, Pb (200ppb) removal from 25 ml aqueous solution using FSBC over 1h.

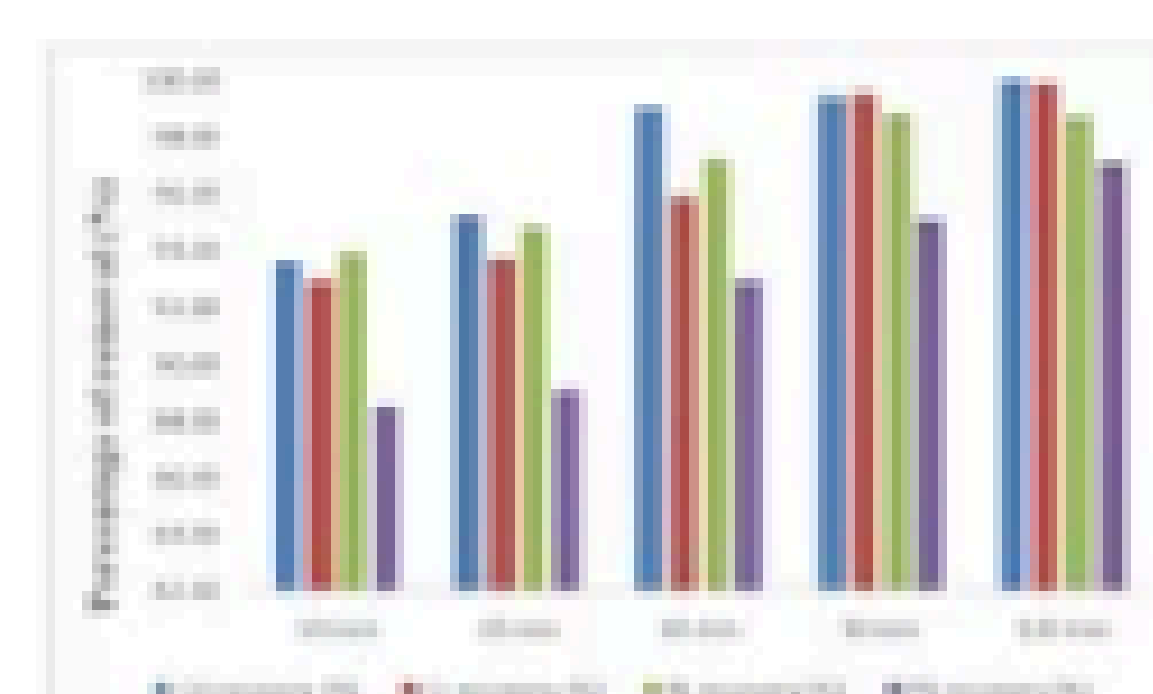


Fig 10. Effect of contact time on Cd, Cr, Ni, and Pb (200ppb) removal from 25 ml aqueous solution using FSBC (25 mg).

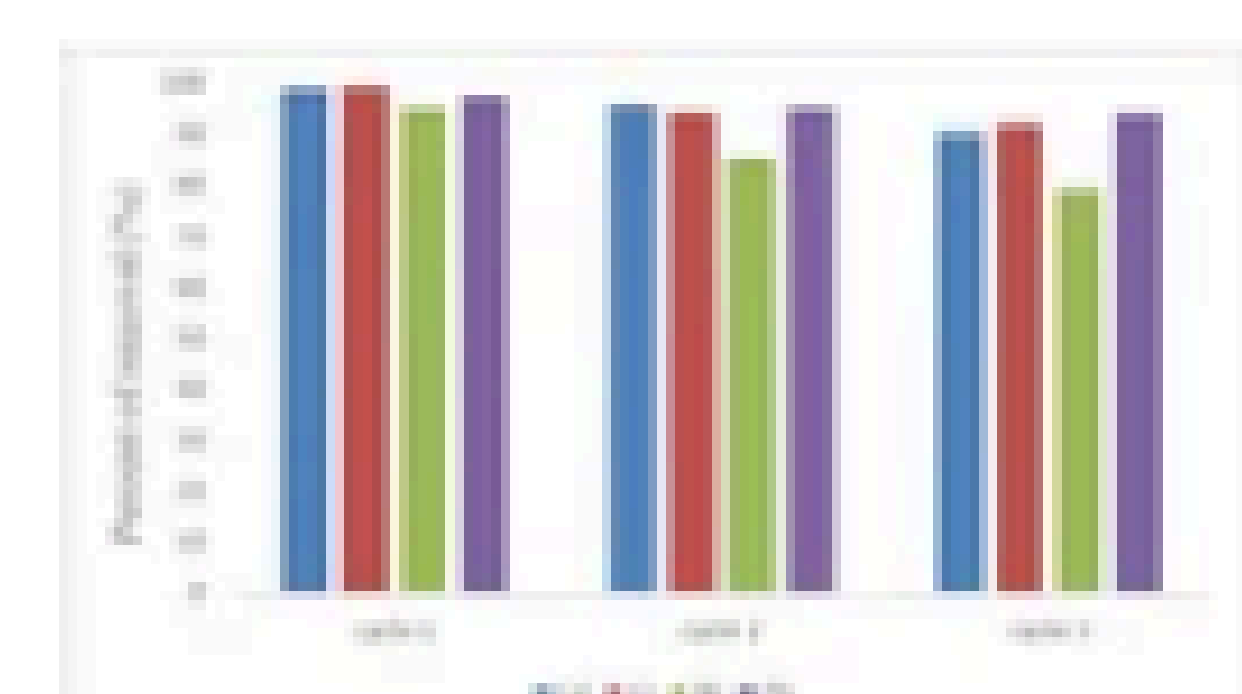


Fig 11. Heavy metal removal efficiency of FSBC over three cycles.

Conclusion

- The experimental results demonstrate that fish biochar-based sorbent is an effective material for the simultaneous removal of heavy metals such as Cd, Cr, Ni, and Pb from aqueous solutions. The sorbent can be easily synthesized from fish waste through a simple process involving washing, drying, grinding, and pyrolysis, making it a low-cost and renewable option suitable for resource-limited settings. Optimal removal efficiency was observed at pH 6, using a sorbent dosage of 25 mg and a contact time of 60 minutes. Furthermore, the sorbent proved effective in both drinking and tap water, highlighting its potential application in continuous water treatment systems.

Future Work

Enhancing the performance and applicability of FSBC-based sorbents through targeted material modifications.

Potential directions include:

- Surface functionalization.
- Magnetic modification.
- Composite material development.
- Metal recovery techniques.
- Scale-up and process integration.



References

